

# openGUTS Report

**Project:**

GUTS\_RF\_Chironomus\_81\_Noprefitbackgroundhazard\_29052026

**Project file:**

No project file saved or loaded

**Project description (optional):**

No project description available

**Software version:**

openGUTS - 1.2

**Date of report creation:**

01/06/2026 11:41:33

# Calibration

## Calibration input data

### Data set 1

File: Data edited manually

Description (optional):

Control group: 'Control'

### Survival data of input data set 1:

| Time [d] | Control | T1 | T2 | T3 | T4 | T5 |
|----------|---------|----|----|----|----|----|
| 0        | 80      | 20 | 20 | 20 | 20 | 20 |
| 0.042    | 80      | 20 | 20 | 20 | 20 | 20 |
| 0.083    | 80      | 20 | 20 | 20 | 20 | 20 |
| 0.16     | 80      | 20 | 20 | 20 | 20 | 20 |
| 0.33     | 80      | 20 | 20 | 20 | 20 | 19 |
| 1        | 80      | 20 | 20 | 20 | 20 | 19 |
| 1.21     | 80      | 20 | 20 | 20 | 20 | 19 |
| 2.083    | 80      | 20 | 20 | 20 | 18 | 18 |
| 3        | 77      | 19 | 19 | 20 | 13 | 15 |
| 4        | 77      | 19 | 19 | 19 | 11 | 10 |

### Concentration data of input data set 1:

| Time [d] | Control | T1   | T2   | T3    | T4    | T5     |
|----------|---------|------|------|-------|-------|--------|
| 0        | 0       | 30   | 100  | 300   | 1000  | 3000   |
| 0.2      | 0       | 29.7 | 99   | 296.9 | 989.5 | 2968.6 |
| 0.4      | 0       | 29.4 | 97.9 | 293.7 | 979.1 | 2937.4 |
| 0.6      | 0       | 29.1 | 96.9 | 290.7 | 968.9 | 2906.6 |
| 0.8      | 0       | 28.8 | 95.9 | 287.6 | 958.7 | 2876.2 |
| 1        | 0       | 28.5 | 94.9 | 284.6 | 948.7 | 2846   |
| 1.2      | 0       | 28.2 | 93.9 | 281.6 | 938.7 | 2816.2 |
| 1.4      | 0       | 27.9 | 92.9 | 278.7 | 928.9 | 2786.7 |
| 1.6      | 0       | 27.6 | 91.9 | 275.7 | 919.2 | 2757.5 |

|     |   |      |      |       |       |        |
|-----|---|------|------|-------|-------|--------|
| 1.8 | 0 | 27.3 | 91   | 272.9 | 909.5 | 2728.5 |
| 2   | 0 | 27   | 90   | 270   | 900   | 2699.9 |
| 2.2 | 0 | 26.7 | 89.1 | 267.2 | 890.5 | 2671.6 |
| 2.4 | 0 | 26.4 | 88.1 | 264.4 | 881.2 | 2643.6 |
| 2.6 | 0 | 26.2 | 87.2 | 261.6 | 872   | 2615.9 |
| 2.8 | 0 | 25.9 | 86.3 | 258.9 | 862.8 | 2588.5 |
| 3   | 0 | 25.6 | 85.4 | 256.1 | 853.8 | 2561.4 |
| 3.2 | 0 | 25.3 | 84.5 | 253.5 | 844.8 | 2534.5 |
| 3.4 | 0 | 25.1 | 83.6 | 250.8 | 836   | 2508   |
| 3.6 | 0 | 24.8 | 82.7 | 248.2 | 827.2 | 2481.7 |
| 3.8 | 0 | 24.6 | 81.9 | 245.6 | 818.5 | 2455.6 |
| 4   | 0 | 24.3 | 81   | 243   | 810   | 2429.9 |

## Calibration settings

Calibration parameter settings for GUTS-RED-SD:

| Parameter | Fit | Min      | Max   | Scale |
|-----------|-----|----------|-------|-------|
| kd        | Yes | 0.001641 | 1096  | Log   |
| mw        | Yes | 0.007403 | 2970  | Norm  |
| hb        | Yes | 1E-6     | 0.07  | Norm  |
| bw        | Yes | 8.78E-6  | 111.5 | Log   |
| Fs        | No  | 1        | 1     | Norm  |

Calibration parameter settings for GUTS-RED-IT:

| Parameter | Fit | Min      | Max  | Scale |
|-----------|-----|----------|------|-------|
| kd        | Yes | 0.001641 | 2.74 | Log   |
| mw        | Yes | 0.007403 | 5834 | Log   |
| hb        | Yes | 1E-6     | 0.07 | Norm  |
| bw        | No  | Inf      | Inf  | Norm  |
| Fs        | Yes | 1.05     | 20   | Log   |

## Calibration results

### Fitted parameters for GUTS-RED-SD:

Best fit parameter values and their 95% CI

kd: 0.001641 (0.001641\* - 0.7611)  
mw: 0.2034 (0.007403\* - 206.1)  
hb: 0.009274 (0.002432 - 0.02109)  
bw: 0.02369 (9.415E-5 - 0.04063)

\* edge of 95% parameter CI has run into a boundary  
(this may also affect CIs of other parameters)

### Goodness of fit for calibration data (GUTS-RED-SD):

Model efficiency (NSE, r-square): 0.779

Normalised root-means-square error (NRMSE): 3.506 %

Minus log-likelihood (MLL): 89.56

AIC: 187.12

Survival probability prediction error (SPPE) for each treatment:

| Data set | Treatment | Value   |
|----------|-----------|---------|
| 1        | Control   | -0.11 % |
| 1        | T1        | -1.36 % |
| 1        | T2        | -0.14 % |
| 1        | T3        | 5.032 % |
| 1        | T4        | -18.5 % |
| 1        | T5        | 8.804 % |

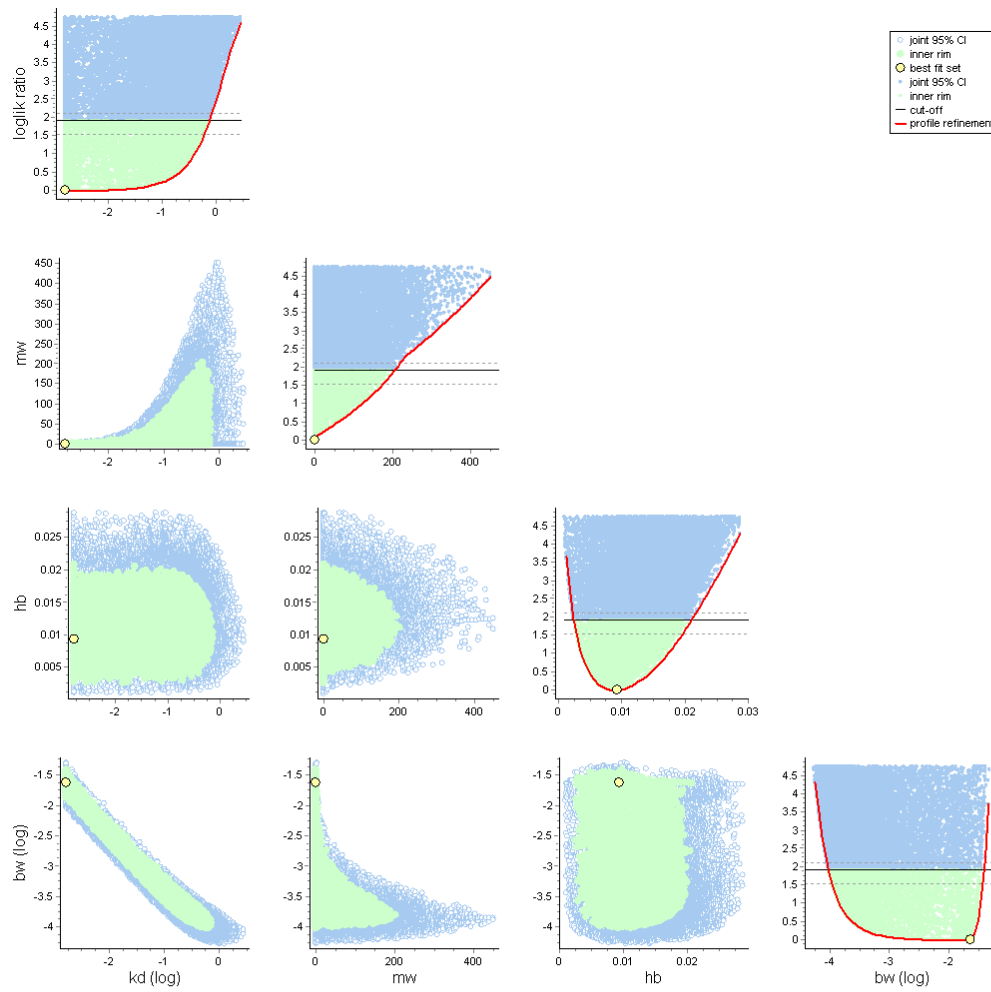
### GUTS-RED-SD results table for LC<sub>x,t</sub> [[C]], with 95% CI:

| Time [d] | LC50                  | LC20                 | LC10                  |
|----------|-----------------------|----------------------|-----------------------|
| 1        | 35925 (17035 - 65398) | 11732 (5850 - 21060) | 5668 (2895 - 9948)    |
| 2        | 9048 (5117 - 16361)   | 2996 (1781 - 5270)   | 1478 (841.4 - 2502)   |
| 3        | 4051 (2568 - 7276)    | 1359 (850.1 - 2345)  | 683.5 (401.4 - 1200)  |
| 4        | 2295 (1485 - 4096)    | 779.9 (479 - 1327)   | 399.2 (226.9 - 751.9) |

|     |                        |                        |                        |
|-----|------------------------|------------------------|------------------------|
| 7   | 765.6 (486.7 - 1506)   | 269.6 (157.2 - 587.3)  | 144.4 (74.61 - 399.9)  |
| 10  | 383.1 (239.3 - 955.9)  | 139.2 (77.4 - 435.2)   | 77.43 (36.82 - 321.7)  |
| 14  | 200.8 (122.7 - 696.5)  | 75.82 (39.75 - 355.5)  | 43.91 (18.96 - 282.5)  |
| 21  | 93.35 (54.94 - 502.5)  | 37.33 (17.87 - 298.7)  | 22.83 (8.566 - 254.6)  |
| 27  | 58.59 (33.42 - 423.1)  | 24.42 (10.91 - 276)    | 15.48 (5.254 - 243.5)  |
| 28  | 54.81 (31.1 - 414.3)   | 22.99 (10.16 - 273.4)  | 14.65 (4.897 - 242.1)  |
| 42  | 26.38 (14 - 337.7)     | 11.95 (4.619 - 251)    | 8.069 (2.247 - 230.8)  |
| 50  | 19.42 (9.952 - 314.7)  | 9.115 (3.301 - 244.2)  | 6.311 (1.614 - 227.4)  |
| 100 | 6.112 (2.602 - 259)    | 3.331 (0.892 - 227.1)  | 2.53 (0.4497 - 219)    |
| 182 | 2.475 (0.8444 - 237.2) | 1.534 (0.3042 - 219.9) | 1.247 (0.1602 - 215.5) |

## Plots for GUTS-RED-SD calibration:

### Parameter space plot for the calibration of GUTS-RED-SD:

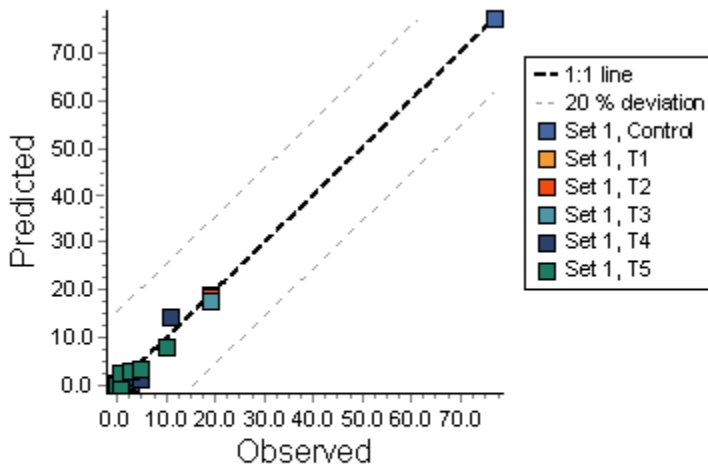


**Exposure, damage and survival plots for the calibration of GUTS-RED-SD:**

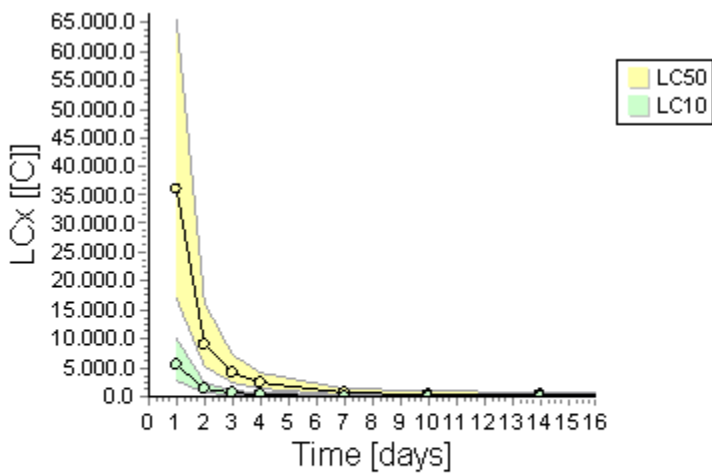
... continued plot:

**Observed vs. Predicted survival plot for the calibration of GUTS-RED-SD:**

**Observed vs. Predicted deaths plot for the calibration of GUTS-RED-SD:**



**LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-SD):**



**Fitted parameters for GUTS-RED-IT:**

Best fit parameter values and their 95% CI

kd: 0.001641 (0.001641\* - 0.1278)

mw: 14.92 (10.19 - 996.8)

hb: 0.01072 (0.003684 - 0.02301)

Fs: 8.611 (4.351 - 20\*)

\* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

**Goodness of fit for calibration data (GUTS-RED-IT):**

Model efficiency (NSE, r-square): 0.5692

Normalised root-means-square error (NRMSE): 4.826 %

Minus log-likelihood (MLL): 93.81

AIC: 195.62

Survival probability prediction error (SPPE) for each treatment:

| Data set | Treatment | Value    |
|----------|-----------|----------|
| 1        | Control   | 0.4471 % |
| 1        | T1        | -0.75 %  |
| 1        | T2        | -0.41 %  |
| 1        | T3        | 1.677 %  |
| 1        | T4        | -24.4 %  |
| 1        | T5        | 8.998 %  |

**GUTS-RED-IT results table for LC<sub>x,t</sub> [[C]], with 95% CI:**

| Time [d] | LC50                 | LC20                  | LC10                  |
|----------|----------------------|-----------------------|-----------------------|
| 1        | 9100 (6089 - 15734)  | 4029 (2487 - 6426)    | 2502 (1338 - 4307)    |
| 2        | 4554 (3048 - 7875)   | 2016 (1246 - 3219)    | 1252 (669.4 - 2157)   |
| 3        | 3038 (2035 - 5256)   | 1345 (832.1 - 2150)   | 835.3 (446.7 - 1443)  |
| 4        | 2281 (1528 - 3946)   | 1010 (625.3 - 1616)   | 627 (335.3 - 1086)    |
| 7        | 1306 (876.4 - 2262)  | 578.5 (359.3 - 928.6) | 359.2 (192.1 - 627.6) |
| 10       | 916.8 (615.6 - 1601) | 405.9 (253 - 715.1)   | 252 (134.8 - 475.1)   |

|     |                      |                       |                       |
|-----|----------------------|-----------------------|-----------------------|
| 14  | 657 (441.2 - 1274)   | 290.9 (182 - 598.1)   | 180.6 (96.62 - 398)   |
| 21  | 440.5 (295.8 - 1107) | 195 (123 - 525.4)     | 121.1 (64.8 - 341)    |
| 27  | 344.3 (231.2 - 1065) | 152.4 (96.44 - 506.7) | 94.65 (50.66 - 328.1) |
| 28  | 332.3 (223.1 - 1061) | 147.1 (93.1 - 504.9)  | 91.35 (48.89 - 327)   |
| 42  | 224 (150.5 - 1039)   | 99.2 (63.08 - 494.1)  | 61.59 (32.98 - 320)   |
| 50  | 189.4 (127.2 - 1036) | 83.87 (53.47 - 492.8) | 52.08 (27.89 - 319.1) |
| 100 | 98.59 (66.21 - 1034) | 43.65 (28.13 - 492.1) | 27.11 (14.54 - 318.7) |
| 182 | 57.79 (38.81 - 1034) | 25.59 (16.49 - 492.1) | 15.89 (8.549 - 318.7) |

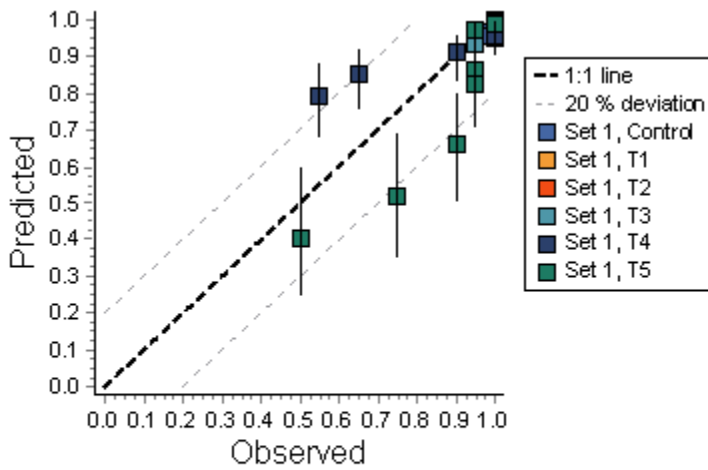
**Plots for GUTS-RED-IT calibration:**

**Parameter space plot for the calibration of GUTS-RED-IT:**

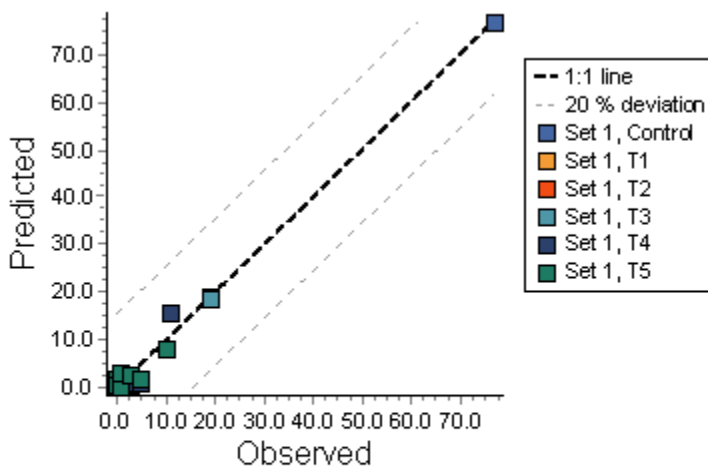
**Exposure, damage and survival plots for the calibration of GUTS-RED-IT:**

... continued plot:

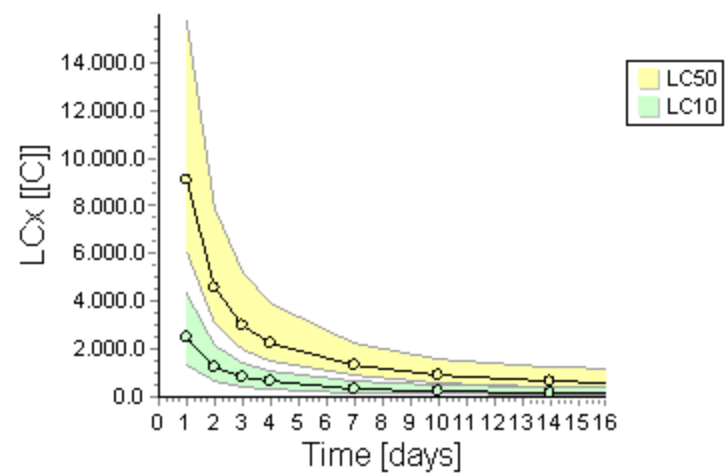
**Observed vs. Predicted survival plot for the calibration of GUTS-RED-IT:**



**Observed vs. Predicted deaths plot for the calibration of GUTS-RED-IT:**



**LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-IT):**



## Validation

No validation performed!

## Predictions

No predictions performed!